Real-Time Semantic Acoustic Imaging via Variational Posterior Tracking and Generative Mixture Fitting *

Bogdan Jastrzębski
Samsung Research Institute Poland (SRPOL)
Warsaw
b.jastrzebsk@samsung.com

Abstract

Current semantic acoustic imaging baselines reconstruct dense 180°×360° spatial maps independently per frame. These systems rely on underdetermined numerical solvers that introduce checkerboard artifacts, forcing the use of early stopping which precludes temporal warm-starting [?]. We propose abandoning dense map regression in favor of a generative Bayesian framework via Variational Inference. We model the unknown true sound sources as a set of points, and the observed audio-visual spatial data as generated by a Gaussian Mixture Model (GMM) conditioned on these points. Instead of tracking rigid point estimates across time, we fit a variational approximate posterior for the current frame, and propagate this full probability distribution to the next frame to serve as its prior. This retains spatial uncertainty across time, achieving fluid real-time temporal coherence while analytically circumventing the artifacts of dense numerical solvers [?].

Keywords Sound Event Localization and Detection · Acoustic Imaging · Variational Inference · Gaussian Mixture Model · Semantic Audio

1 Introduction

Because baseline pipelines map 32 microphones onto 484 spatial directions, the mathematical operator is fundamentally singular [?]. Exact solvers like ADMM converge to minimizers filled with ringing artifacts, requiring early stopping to produce smooth approximations [?]. Consequently, current models cannot use the solution from time $t-1$ to initialize time t, because warm-starting accelerates convergence toward the artifact-ridden exact minimizer [?]. We bypass this bottleneck by modeling the scene as a sparse, unknown set of point sources and fitting an approximate posterior distribution over their locations.

2 Probabilistic Generative Model

Let $\mathbf{P}^{(t)} = \{p_1^{(t)}, \dots, p_K^{(t)}\}$ represent the unknown spatial locations of K sound sources at time t. The observed spatial features extracted from the audio and visual front-ends are represented as data points $\mathbf{D}^{(t)} = \{x_1, \dots, x_N\}$.

2.1 The Likelihood (Observation Model)

We assume the sound diffuses spatially from the true points $\mathbf{P}^{(t)}$, generating observations according to a Gaussian Mixture Model:

$$p(\mathbf{D}^{(t)} \mid \mathbf{P}^{(t)}) = \prod_{i=1}^{N} \left(\sum_{k=1}^{K} \pi_k \mathcal{N}(x_i \mid p_k^{(t)}, \Sigma_A) + \pi_0 \mathcal{U}(x_i) \right) \tag{1}$$

where Σ_A represents the inherent spatial spread of the acoustic energy.

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2.2 The Temporal Prior

The true point locations transition smoothly over time. The prior for the current frame t is formed by marginalizing over the transition dynamics from the previous frame's posterior:

$$p(\mathbf{P}^{(t)} \mid \mathbf{D}_{1:t-1}) = \int p(\mathbf{P}^{(t)} \mid \mathbf{P}^{(t-1)}) p(\mathbf{P}^{(t-1)} \mid \mathbf{D}_{1:t-1})\, d\mathbf{P}^{(t-1)} \tag{2}$$

3 Variational Inference and Posterior Tracking

Computing the exact posterior $p(\mathbf{P}^{(t)} \mid \mathbf{D}_{1:t})$ is intractable. Instead, we introduce a tractable variational distribution $q^{(t)}(\mathbf{P})$ to approximate the true posterior. We parameterize $q^{(t)}$ as a set of independent Gaussians over the point locations:

$$q^{(t)}(\mathbf{P}) = \prod_{k=1}^{K} \mathcal{N}\left(p_k \mid \mu_k^{(t)}, \Sigma_k^{(t)}\right) \tag{3}$$

Here, $\mu_k^{(t)}$ is the model's belief about the location of source k, and $\Sigma_k^{(t)}$ encapsulates the uncertainty of that belief.

3.1 Optimizing the ELBO

To fit this approximate posterior, we maximize the Evidence Lower Bound (ELBO):

$$\mathcal{L}(q^{(t)}) = \mathbb{E}_{q^{(t)}}\left[\log p(\mathbf{D}^{(t)} \mid \mathbf{P}^{(t)})\right] - \mathrm{KL}\left(q^{(t)}(\mathbf{P}) \parallel p(\mathbf{P}^{(t)} \mid \mathbf{D}_{1:t-1})\right) \tag{4}$$

The first term forces the approximate posterior to cover regions where the observed audio-visual data is dense (maximizing coverage). The KL-divergence term acts as a temporal regularizer, penalizing the distribution if it deviates too far from the prior established by the previous frame.

3.2 State Propagation and Prediction

Once the ELBO is maximized for frame t, we obtain our fitted approximate posterior $q^{(t)}(\mathbf{P})$. For the final semantic map prediction required by the evaluator, we take the MAP estimate of this distribution:

$$\hat{\mathbf{P}}^{(t)} = \arg\max_{\mathbf{P}} q^{(t)}(\mathbf{P}) = \{\mu_1^{(t)}, \ldots, \mu_K^{(t)}\} \tag{5}$$

Crucially, the entire distribution $q^{(t)}(\mathbf{P})$ (including the variance $\Sigma_k^{(t)}$) is stored in memory and propagated to frame $t+1$ to serve as the new prior, preserving the temporal accumulation of uncertainty and enabling highly efficient, artifact-free tracking.

4 Conclusion

By formulating SELD as a continuous Bayesian state-estimation problem with full variational posterior tracking, the model propagates uncertainty across time instead of discarding it. This analytically circumvents checkerboard artifacts of exact numerical solvers [**?**] and enables fluid real-time temporal coherence by warm-starting each frame with the previous posterior distribution.

Acknowledgments

References